ELECTRICAL ENERGY AUDITING OF BIT MESRA OFF-CAMPUS PATNA

A Thesis

Submitted in partial fulfillment of the
Requirements for the award of the degree of

BACHELOR OF TECHNOLOGY
IN
ELECTRICAL & ELECTRONICS ENGINEERING

BY

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APPROVAL OF THE GUIDE

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- The work has not been submitted to any other Institute for any degree or diploma.
- We have followed the guidelines provided by the Institute in writing the project report.
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ABSTRACT

This paper analyses the electrical energy audit carried out in BIT Mesra Off-Campus, Patna. The primary purpose of performing such an audit was to examine the existing usage of energy and find any possibilities for energy reduction without disturbing the normal functioning of the business operations.

During the auditing process, different aspects of the facility, including the usage in hostels, academic buildings, street lightings and utilities zones, were considered. It has been found that some machines were consuming relatively higher amounts of electricity than initially expected due to the longer duration of operation in some cases.

During the audit process, different methods that could be used to increase the level of energy efficiency of the system were also examined. For example, installation of LEDs instead of standard lights, setting up solar street lightings and usage of solar water heaters were among the options found during the auditing process due to their relative ease and ability to conserve energy.

Finally, practical recommendations were developed based on the outcomes of the analysis conducted. They include usage of LED lights, solar street lights and replacement of existing electric water heaters with solar ones. Evaluation of these recommendations included technical and economic criteria – namely how fast would they provide initial investment recovery.

It is clear that there are certain possibilities for reduction of energy consumption level in BIT Mesra Off-Campus, Patna, which does not involve disturbance of any ongoing business processes. In addition to cost-effectiveness, other benefits of such actions include environmental protection.

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LIST OF ABBREVIATIONS

- HVAC – Heating, Ventilation, and Air Conditioning
- CFL – Compact Fluorescent Lamp
- LED – Light Emitting Diode
- FPC – Flat Plate Collector
- ETC – Evacuated Tube Collector
- BSEB – Bihar State Electricity Board
- BIT – Birla Institute of Technology

CHAPTER 1: INTRODUCTION

1.1 Introduction

Electrical energy is among the most significant resources in any educational establishment. On a college or university campus, electrical energy is used virtually everywhere – lecture halls, labs, dormitories, water supply, lighting systems, and even smaller buildings such as cafeterias or stores. Therefore, total energy consumption is inevitably quite high.

In simple terms, an energy audit is a procedure that enables you to understand the consumption of this electricity. However, it does not only mean that you look through your electricity bills but also that you examine your energy consumption in detail to see whether any of it goes to waste. Often, there is no single point of energy loss but a series of minor losses that are usually overlooked.

There is a general procedure for conducting an energy audit. You have to collect the information regarding electricity bills, inspect the power rating of your appliances, and observe for how long each system works. At the same time, a physical examination should be conducted since practical usage always varies from expected usage.

After thorough investigation, some improvements could be recommended. They do not necessarily mean decreasing your energy consumption but rather preventing unnecessary expenses.

Due to rising energy prices and growing environmental awareness, the issue of responsible energy usage has been gaining more attention. Institutions of such a scale as BIT Mesra can benefit significantly from implementing efficient solutions.

The energy auditing process starts by selecting people who will carry out the audit and creating a team of specialists responsible for conducting inspections and gathering information. Every team member should select his or her site for analysis and determine energy consumption in it by using suitable instruments. Each team member then collects the necessary data for auditing purposes. The energy auditing process includes three major stages: energy survey and analysis and detailed analysis. Finally, collected data are analyzed according to these three stages to improve energy efficiency and evaluate the outcomes. Recommendations are provided after every stage and serve for energy conservation.

This report concentrates on analyzing the energy consumption of the Patna campus and identifying possible areas for improvement.

1.2 Recognition of Need

With the growth in recent years in energy consumption because of the infrastructural growth and dependency on the electric system, education facilities have been included in this list because of their prolonged functioning times. Energy consumption has not been optimized in various cases. There can be excessive energy utilization in unnecessary operations and old technological devices, which could have better replacements. Even though each case might be insignificant alone, the cumulative effect could lead to high energy consumption.

The first benefit of conducting an energy audit includes pinpointing the areas where optimization can occur. An audit will provide more information on how much energy is utilized and how to reduce its consumption. This will be beneficial because the problem area would be identified, and there would be room for action.

There is also another advantage concerning the impact of the environment because of carbon emissions from the consumption of energy. Electricity consumption will generally lead to carbon emission if it comes from fossil fuel sources. Reduction in energy consumption can lead to lower environmental harm caused by the education facilities.

There is another advantage concerning the usability of the facilities. Optimization of energy consumption will also improve the overall quality of the facility, including lighting and ventilation. Improvements will also enhance the usability of the building.

For all these reasons, it becomes clear that the necessity for conducting energy audits is becoming increasingly significant.

CHAPTER 2: LITERATURE REVIEW

2.1 Literature Survey

Researches on energy audits have covered various facilities such as industries, commercial spaces and educational establishments. Majority of them provide recommendations for improvement which are technically possible as well as cost-effective.

Among other methodologies, case study is quite popular in energy audit-related literature. An example of case study methodology implementation includes a research on energy audit in an Indian industry. It is noted in the research that accurate measurements should be taken into account as opposed to estimations. As a result of the research, the researchers noticed that considerable amounts of energy were being used by the machinery in their idle state. After the detection of the problem, minor operational adjustments were performed which resulted in the reduction of energy use. This implies that there is no need for expensive changes as far as saving energy is concerned since sometimes it is possible to reach good results by making sure of effective energy consumption.

The second example refers to a research on the use of energy in a college building. Various components were audited including lighting, air conditioning, laboratory equipment. Among other observations, it was noted that lighting is responsible for considerable consumption rates. It was possible to reduce overall energy costs after switching from fluorescent lamps to LEDs without altering the illumination levels. As it may be seen from the provided case study, it was possible to obtain tangible results through implementation of the simplest solution.

There have been numerous examples showing how it is possible to reduce energy consumption in school settings through application of different methods. For example, energy audit in a school building demonstrated that equipment such as lamps and fans was kept running although they were unnecessary. As a result, after simple adjustment procedures and education activities, it was possible to reduce unnecessary consumption. As one might see from the provided cases, the human factor is quite significant.

Apart from lighting and use of equipment, another common area covered by literature includes heating and cooling systems. In areas with very severe climatic conditions, these systems consume a very large amount of energy. According to studies, efficient use of energy can only come about through regulation of the system's temperature and proper maintenance. It is even possible for institutions to install automated control systems that operate based on whether or not there are occupants.

Some literature has shown that universities that do not lie within the borders of India have different experiences altogether when it comes to energy audit. In such universities, management of energy is viewed as a process and not as a one-off activity. Some universities have managed to install centralized monitoring systems that give information about energy usage at any point in time. Although these systems may require a lot of money to install, the long-term benefits are immense.

One aspect that is extensively covered in the literature is how renewable energy resources have been incorporated in institutional energy audits. This topic has received much attention because solar energy is quite easy to harvest. In most literature, rooftop solar panels have been installed in institutions to provide supplemental power to the grid. According to studies, although the capital cost is very high, there are substantial gains that come with using this resource in the long run.

Another form of renewable energy that is discussed extensively is the solar water heating system. This is particularly useful in institutions that occupy hostels and other residential quarters. Literature has revealed that the replacement of electric geysers with solar heaters has yielded great results in minimizing energy consumption. However, performance of such a system is highly dependent on weather conditions and proper maintenance.

The success that energy audit yields in various studies varies greatly depending on many factors. However, it is generally possible to conclude that successful energy audits are based on four stages – data collection, data analysis, problem identification, and recommendation. Nonetheless, the success of these audits depends on how accurately data is collected and recommendations made.

It should also be acknowledged that not all recommendations are implemented instantly. In general, organizations tend to prefer cost-effective measures first of all, including turning off all unnecessary equipment or using energy-efficient lighting. Other changes requiring large investments, for instance, implementing renewable energy systems, will follow if there is enough money.

A third finding, made according to the literature analysis, is that energy audits become more productive when accompanied by user awareness activities. As people understand the influence their behavior has, they are more likely to engage in energy efficient activities. In addition, some organizations include students in their efforts to monitor energy consumption and create user awareness.

At the same time, there are some limitations related to conducting energy audits that need to be addressed. First of all, energy efficiency depends to a great extent on the season when audit is performed. Thus, for example, consumption in summer is expected to differ greatly from winter because of cooling needs.

Moreover, lack of maintenance proved to be a common problem of energy efficiency, leading to energy wastage. Equipment that is not maintained properly works poorly and consumes more power. Despite the fact that maintenance is not usually seen as crucial, it plays an extremely important role.

Summarizing the findings of all reviewed literature, it should be said that energy auditing is not a one-time activity. Though initial audits give organizations insight into the most important problems, further monitoring is needed for energy efficiency to remain high.

Taking into account the above information about different aspects of energy audit, one can state for sure that energy savings are possible due to various actions, both technical and organizational, as well as creating awareness among users. The conclusions from the literature review will serve as the basis for further investigation.

CHAPTER 3: ENERGY AUDIT AND METHODOLOGY

3.1 Project Objective

The main projectiles of this project stand as:

- To materialize current energy consumption pattern of the campus
- To identify areas where energy usage is inefficient
- To analyze the feasibility of renewable energy solutions
- To suggest practical measures for reducing energy consumption

3.2 Energy Audit

Energy audits offer us the greatest potential to have reduced costs, making the energy audit a vital exercise. Comprehensive energy audits serve as a diagnostic tool for the sector, revealing exactly where power is being consumed and, more importantly, where it is being lost. This process effectively bridges the gap between current operational waste and potential efficiency gains. These energy audit rules require a few key steps. First, you gather data on current energy use. Next, you do a deep check of the usage profile to find specific trends. Finally, when looking at costs, you must calculate the expense over the whole life of the project rather than just using a simple payback period.

Energy Audit - Need:

The need for energy is growing every day. More fossil fuels must be used in order to meet this demand. Fossil fuels produce pollution that is bad for humanity as well as energy. On the internet, there are calculators for checking emissions and pollution that are used to help measure Environmental contamination and emission levels. An energy audit will either directly or indirectly lessen the impact on the environment. An energy audit has the following benefits:

1. it can lower energy use
2. it can lower energy costs and save money
3. it can raise comfort levels
4. it can lower carbon footprints
5. it can lower wasteful waste and pollution

The best and most effective way to reduce energy use and costs is to conduct an energy audit. It may appear that energy audit an organization gives us a point of reference for energy usage optimization as well as a better strategy for using energy in an organization's core operations

Types of Energy Auditing:

Energy audit is the greatest way to lower energy bills and energy usage, no matter being it is a commercial or non–commercial sector. The kind of energy audit that must be carried out based on:

1. the industry in question, its kind.
2. Extent of the audit, and the accumulated data amount.
3. Transparency and approval needed for data collection is important.

The three types of energy audit are:

1. Walk-through energy audit
2. Target energy audit
3. Detailed energy audit

3.2.1. Walk through energy audit

For a walkthrough of an energy review is a fundamental overview directed by inspectors to explore an association's energy utilization. Dissimilar to other energy reviews, this kind of review doesn't include significant stages to decrease energy utilization and is frequently alluded to as a fundamental review. It is the first and least complex sort of review that spotlights, recognizing the fundamental areas of energy utilization. The review is finished rapidly and effectively, with the ensured energy supervisor, confirmed evaluator, or expert specialist fostering a thorough rundown of energy-saving measures. In any case, the subtleties acquired from the walkthrough energy review may not be adequate to recognize energy-saving arrangements. Regardless, the reviewer can make a powerful move in a more limited time with more prominent precision and benefit. The walkthrough energy review furnishes evaluators with every one of the important subtleties of an association, which can be utilized for a definite energy review. The essential extent of walkthrough of energy review in question is to decide the optimal energy usage of an institution.

1. During this kind of review, just a negligible meeting with the pertinent faculty in the association is led.
2. The review essentially centers around distinguishing and resolving the serious trouble spots.
3. To reduce consumption and increase efficiency
4. Cost cuttings despite the improvements

3.2.2 Targeted energy audit

Objective energy review purely depends on the discoveries made in duration of the walkthrough energy review and gives a more definite examination of a particular undertaking. The methodology for this kind of review might differ from relying upon the sort of construction, whether it is a business, private, or modern structure, yet it generally requires more extensive data and information. The result of this review is a bunch of proposals for the moves to be made, which ought to be introduced in a reasonable and succinct energy report that incorporates an activity plan for further developing energy effectiveness. The extent of an objective energy review incorporates:

1. Laying out the energy utilization of the association
2. Directing a negligible meeting with pertinent faculty in the association
3. Zeroing in exclusively on the designated region during the review
4. Recognizing the best and attainable ways of further developing productivity estimations and diminish energy utilization, it is least demanding and generally effective to focus on those that.
5. Distinguishing likely enhancements and reserve funds, especially those that are minimal expense or have an exceptional yield on speculation

The detailed energy audit is otherwise called a venture grade or thorough energy audit. It includes a total investigation of conceivable capital-concentrated changes, including demonstrating and reenactment. This kind of audit might require information assortment over a drawn-out period to guarantee high precision, and it gives definite reports on energy utilization and investment funds. This degree of detail makes it more straightforward for workers for hire to comprehend the particular estimates that should be introduced. The definite energy audit is led in three foresaid stages: the pre-audit stage, the audit stage, and the post-audit stage

- **Pre-audit phase**

The pre-audit stage is the underlying move toward leading a complete energy audit. During this stage, the energy examiner arranges and anticipates the audit. The examiner conducts formal meetings with the President, plant administrator, energy chief, and creation director, as well as brief gatherings with all pertinent faculty. The energy inspector or licensed energy evaluator expects to go to the accompanying lengths during the pre-audit stage:

1. Talking about the reason for the far-reaching energy audit with the site chief
2. Making sense of the significance of all the data from the complete energy audit and planning for the information required during the audit stage
3. Breaking down the significant areas of energy utilization
4. Getting site drawings, for example, a solitary line chart of the electrical circuit, building design, and air conditioning framework
5. The results of this stage ought to assist with settling the audit group, comprehend the board's assumptions for the extensive energy audit, and plan the whole audit inside a particular time span.

- **Audit phase**

Audit stage is comes as next step of an exhaustive process. It is subject to transparency and intricacy of the location being audited. This auditing stage may span from days to weeks or even months for instance and may incorporate non-working hours and evenings to guarantee that nothing is disregarded. During this stage, the accompanying data is gathered:

1. Energy supply data
2. Energy bills for tax and cost information
3. Load area information
4. Audit of current energy the executive's methodology and preparing programs
5. Energy stream outline
6. The gathered information is utilized to:
7. Plan process stream graphs and energy and material adjust
8. Distinguish open doors for energy preservation (ENCON)
9. Make proposals for energy preservation choices
10. Decide energy investment funds and restitution periods
11. Get ready specialized and possibility reports
12. Foster a plan for optimization of energy usage measures

- **Post audit phase**

To evaluate the viability of the energy-saving estimates executed in the past stages, a post-audit stage is required. The post-audit stage includes the accompanying techniques:

1. Carrying out the ENCON proposals and measures
2. Checking the presentation of the executed measures
3. Assessing the aftereffects of the carried-out measures
4. Making remedial moves if essential

The objective of the post-audit stage is to guarantee that the carried-out measures are giving the normal energy reserve funds and to distinguish any regions that might require further improvement. It is a fundamental stage in the energy audit cycle to accomplish the ideal energy productivity and cost investment funds for an association.

3.3 Methodology

The energy auditing cycle can be directed utilizing explicit rules. The energy evaluator should have starter information on the office's energy utilization limit and record specialized subtleties like site drawings, gear stock, and cycle charts. The cycle typically includes four huge advances: arranging, maintaining, examining, and carrying out. The progress of energy audit relies upon great preparation, data gathering, estimation of energy utilization, improvement of various protection techniques, and determination of financially savvy smart courses of action. It is fundamental for structure an energy audit group with sufficient information and abilities related with building gear and frameworks. In this review, information gathered, audited, and dissected to accomplish research objectives. The information gathered through private meetings with important people answerable for the school's foundation, activity, and support. Month to month power bills throughout recent years were additionally gotten. Various field visits to the establishment were directed, and significant perceptions were recorded. During the visits, data with respect to energy-consuming gear, including brands, models, and working hours, was surveyed. The way of behaving of staff, understudies, and managers nearby was addition.

CHAPTER 4: AREA SURVEY AND CALCULATIONS

4.1 Overview of Existing Building

The BIT Mesra Patna campus first opened its doors in 2006. Initially, the academic work at the Patna center began in September 2006 with the first B.E. admissions. These early programs focused on Electronics, Computer Science, and Electrical engineering. Now in 2026, the school provides BTech and MTech degrees in areas like Information Technology, Civil, and Mechanical Engineering, as well as Electrical and Electronics. It also brings in students for BCA, BBA, and MBA courses.

In total, the campus area is covered with 25 acres of land. Out of the 25 acres, 5.25 lakh square feet area is covered by the academic blocks of the institution. There are enough accommodations for the hostel, staff accommodation, clinics, playgrounds, and gymnasium. Apart from that, there are other facilities such as the cafeteria, general store, and guest house at the campus too. The residential center of Patna is a fully equipped facility where all the students stay. Each hostel has large dinning halls where students can have their food.

The overall appearance of the campus along with the facilities ensures a good atmosphere for studies. The classes have state-of-the-art infrastructure with Internet access, as well as Wi-Fi and Audio/Visual aids. The infrastructure of the institute has modular labs, workshops, and computer labs equipped with latest software. The central library consists of more than 15000 books and several research journals. The institute also provides large lecture halls that provide an excellent learning experience. The source of energy on campus is the Bihar State Electricity Board (BSEB). There is also a medical facility in the campus that serves all the health-related purposes of the individuals on campus. Separate hostels are there for the boys and girls that are well equipped with furniture and gaming areas.

BIT Patna has updated electrical systems that provide steady power to the whole campus. To make sure there are no cuts, the campus has a 1MW captive power plant. This ensures that the power supply stays on for the entire campus at all times.

The campus also has well established electrical network that includes transformers, switch gears, and distribution panels. The electrical network is designed to cater to the power requirements of the campus and is regularly maintained and upgraded to ensure smooth functioning.

Overall, BIT Mesra Patna Campus has robust electrical facilities that provide reliable and sustainable power supply to the campus. The campus is committed to promoting energy efficiency and sustainability and is continuously striving to adopt new technologies and practices to achieve this goal.

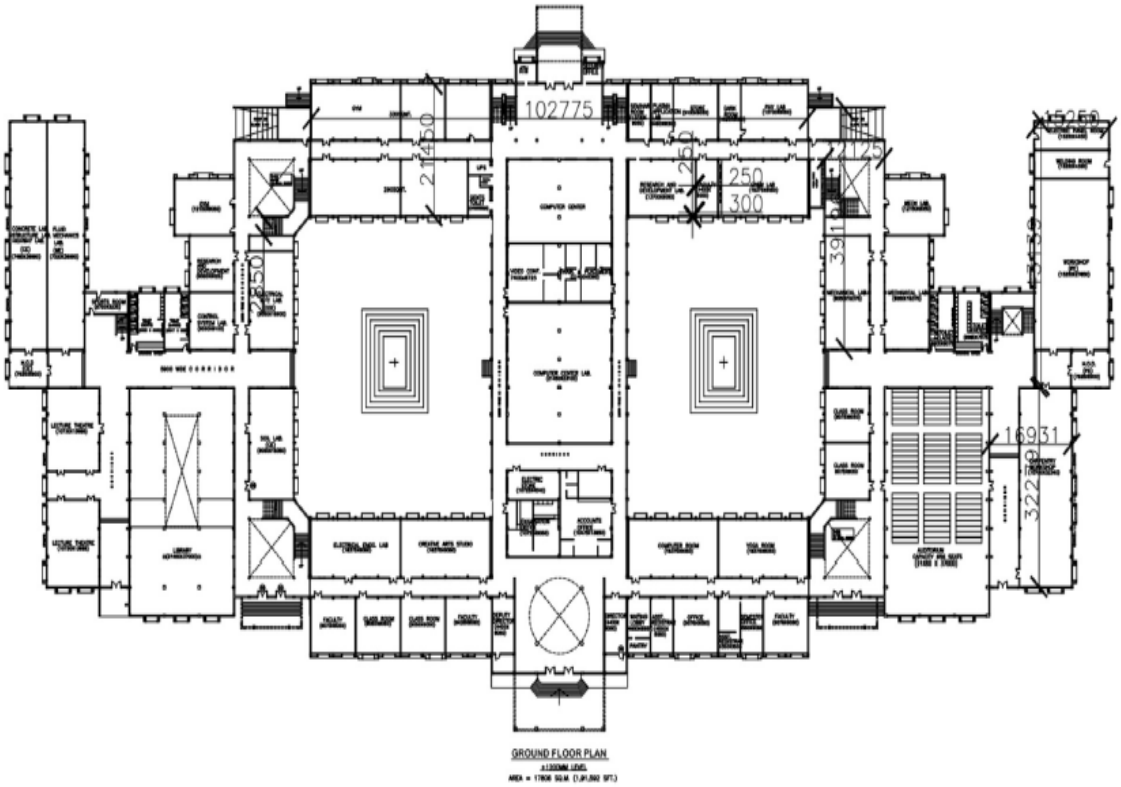


Fig 4.1 layout of ground floor of the institute

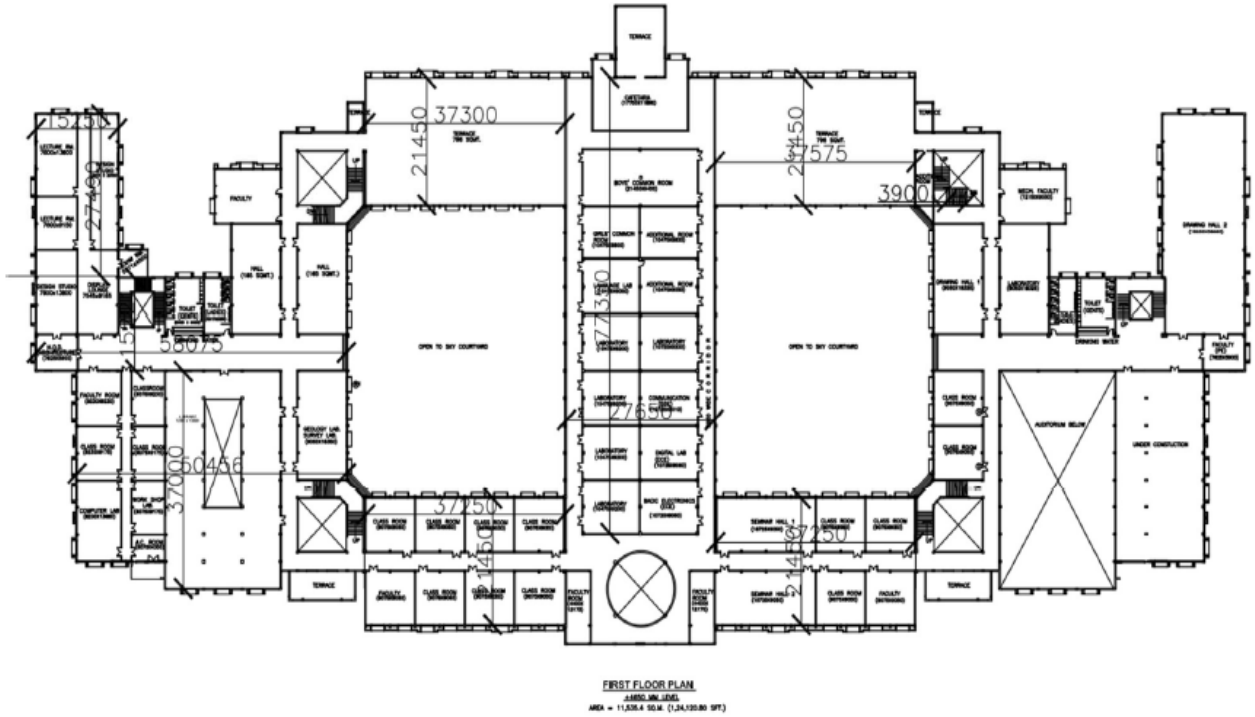


Fig 4.2: layout of first floor of the institute

4.2 Energy Distribution Systems

BIT Patna gets its electricity from Bihar State Electricity Board (BSEB) by two step down transformers, one of 200 KVA and another of 450 KVA i.e. two incoming incomer 1 and incomer 2.

Incomer 1 goes to BH2, BH3, Staff Quarters, library, mechanical lab, guest house, auditorium, workshop, computer lab 1 and 2.

Incomer 2 goes to police station, Barrak, streetlight, BH1, ATM, GH, dispensary.

LOAD DETAILS

To calculate the campus load, divide the entire area into 5 blocks, as shown below:

Group 1 Pump Room.

Group 2 Faculty Quarter (Block A & Block B).

Group 3 Hostels (B1, B2, B3, GH).

Group 4 Guest House.

Group 5 Street Lights, Court Lights.

Group 6 Guard Room, Security, Gail, General store, Canteen, ATM**,** Nescafe.

Group 7 Academic Block



Fig. 4.3 Google view of the campus

A. **Group 1** Pump room

There are two motors installed for supply of underground water to all parts of college.

One motor of 15 hp is used to supply water to all three boys' hostels, two dining halls, E&H, SPH, MTech and the main academic building.

Another motor of 5 hp is used to supply water to three main buildings (girls hostel, staff quarters, main water pump of college, & new campus supply).

Important Data:

1 HP = 746 watt (Since 1 HP = 746 Watt)

Unit Consumption Calculation of 15 HP Water Pump

15 HP = 15 × 746 = 11190 watt
(For 1 HP is equal to 746 Watt)

Now, the total energy eat up for 13 hr = 13 × 11190 = 145470 Whour
Now, the energy usage for a month accounts to: 30 × 13 × 11190 = 4356000 Whour

Total consumption counts to 4356000 Whour

Knowing that 1 KWh = 1000 Whour

Therefore, total consumption in KWh = 4356000 / 1000 = 4356.0 KWh

We also know that 1 unit equals to 1 KWh

Therefore, total usage in one month = 4356.0 Unit

Unit Consumption Calculation of 5 HP Water Pump

5 HP = 5 × 746 = 3730 watt
(Since 1 HP equals to that of 746 Watt)

The energy usage in 13 hr = 13 × 3730 = 48490 Whour

Therefore energy consumption in a month (that is 30 days) = 30 × 13 × 3730 = 1454700 Whour

Total usage adds upto 1454700 Whour
Whour
We know that 1 KWh = 1000 Whour

Therefore, total consumption in KWh = 1454700 / 1000 = 1454.7 KWh

We also know that 1 unit = 1 kWh

Therefore, total energy utilised in one month = 1454.7 Unit

In mini project, another 20 HP water pump consumption increases due to hot weather; therefore pump working hours increases from 13 hr to 15 hr.

20 HP = 20 × 746 = 14920 watt
(Since 1 HP add upto 746 Watt)

The energy consumption in 15 hr = 15 × 14920 watt = 223800 Whour

The energy used in one month (that is 30 days) = 30 × 15 × 14920 Whour = 6714000 Whour

We know that 1 KWh = 1000 Whour

Therefore, total consumption in kWh = 6714000 / 1000 = 6714 kWh

We also know that 1 unit = 1 kWh

Therefore, total energy units in one month = 6714 Unit

B. **Group 2 :** Faculty Quarter (Block A & Block B)

Unit Calculation of Parking and Stair Light of Faculty Quarter

Block A – 14 Led Bulb (9 watt) + 2 Tube Light (40 watt)

Block B – 16 Led Bulb (9 watt) + 3 Tube Light (40 watt)

Total watt by Led Bulb = 30 × 9 = 270 watt

Total watt by Tube Light = 5 × 40 = 200 watt

Now, the energy consumption in 13 hr = 13 × 470 Whour = 6110 Whour

The energy utilised in one month (that is 30 days) = 30 × 13 × 470 Whour

Total usage is = 183300 Whour

Knowing that 1 kWh = 1000 Whour

Therefore, total consumption in kWh = 183300 / 1000 = 183.3 kWh = 183.3 unit

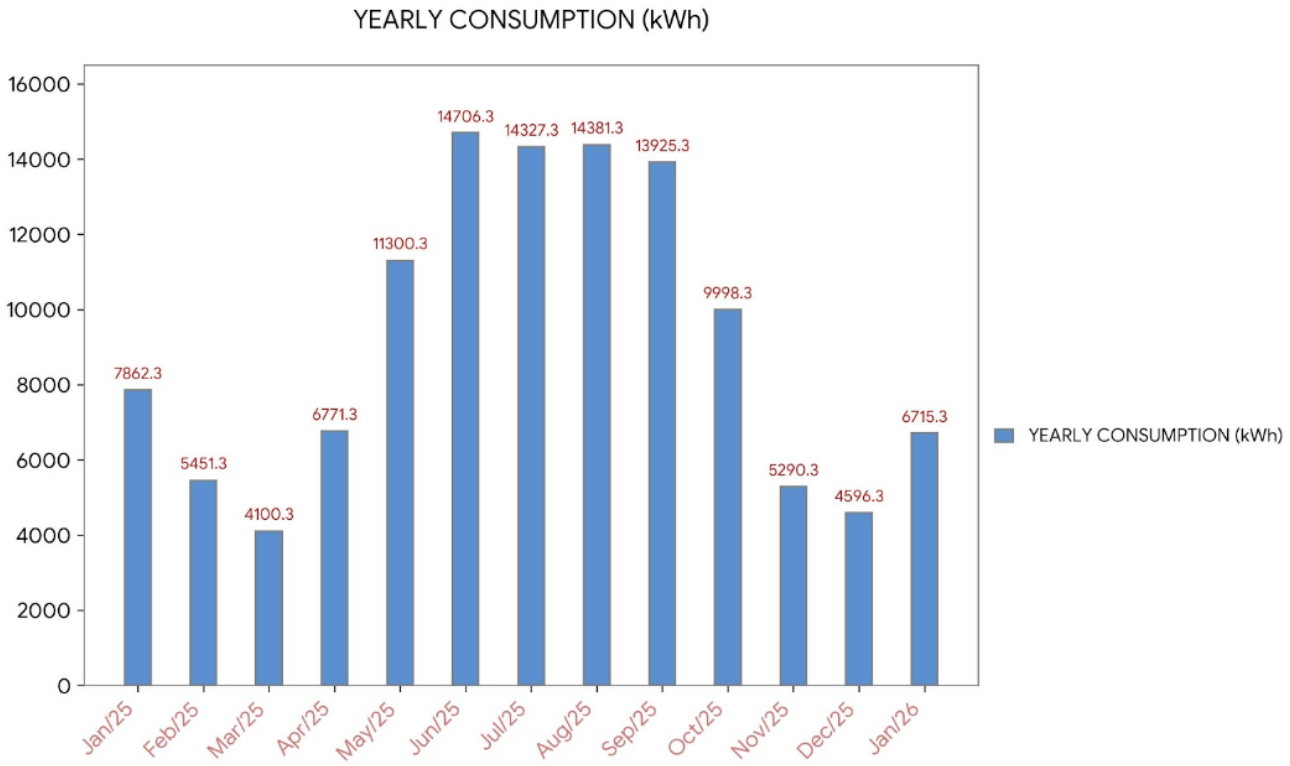


Fig.4.4 Graph of yearly Unit Consumption by Faculty Quarters

Month	Units consumed Block A	Units consumed Block B	Units consumed in stairs and parking	Total Consumption
Jan-25	4130	3549	183.3	7862.3
Feb-25	2945	2323	183.3	5451.3
Mar-25	2135	1782	183.3	4100.3
Apr-25	3227	3361	183.3	6771.3
May-25	5281	5836	183.3	11300.3
Jun-25	6614	7909	183.3	14706.3
Jul-25	7160	6984	183.3	14327.3
Aug-25	6523	7675	183.3	14381.3
Sep-25	6200	7542	183.3	13925.3
Oct-25	4250	5565	183.3	9998.3
Nov-25	2322	2785	183.3	5290.3
Dec-25	1903	2510	183.3	4596.3
Jan-26	2765	3767	183.3	6715.3

Units Consumed by Faculty Quarters

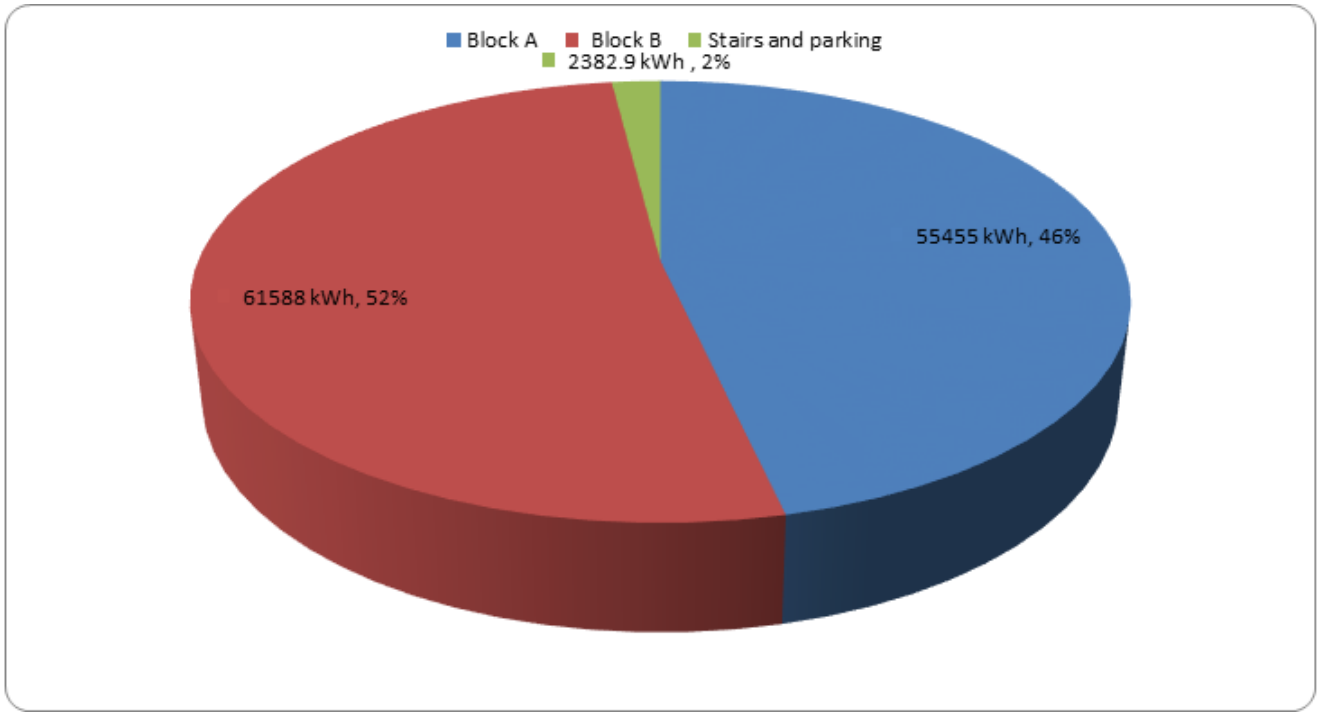


Fig. 4.5 Pie chart indicating Yearly Consumption Division of Faculty quarters

C. **Group 3 :** Hostels

1. **GIRLS HOSTEL**

AREA	LOAD	WATTAGE	CONNECTED LOAD QTY.	TOTAL LOAD CONNECTED (W)	LOAD IN USED (QTY.)
GIRLS HOSTEL	Tubelight	36	220	7920	200
	Fans	60	200	12000	176
	LEDs	9	140	1260	75
	Geysers	2000	19	38000	2
	Water Cooler	1550	3	1650	3
	Water Purifier	35	3	105	3
	Refrigerator	210	1	210	1
	Freezer	1800	1	1800	1
	Grinder	750	1	750	1
	Exhaust	300	3	900	3
	Dough Machine	746	1	746	1
	TV	180	1	180	0

2. **BOYS HOSTEL 1**

AREA	LOAD	WATTAGE	CONNECTED LOAD QTY.	TOTAL LOAD CONNECTED	LOAD IN USE QTY.
BOYS HOSTEL 1	Tubelight	36	318	11448	262
	Fans	60	235	14100	197
	LEDs	9	18	162	16
	Geysers	2000	18	36000	2
	Water Cooler	1550	5	7750	5
	Water Purifier	35	5	175	5
	Refrigerator	210	1	210	1
	Freezer	1800	1	1800	1
	Grinder	750	1	750	1
	Exhaust	300	3	900	3
	Dough Machine	746	1	746	1
	TV	180	1	180	0

3. **BOYS HOSTEL 2**

AREA	LOAD	WATTAGE	CONNECTED LOAD QTY.	TOTAL LOAD CONNECTED	LOAD IN USE QTY.
BOYS HOSTEL 2	Tubelight	36	390	14040	292
	Fans	60	365	21900	312
	LEDs	9	45	405	42
	Geysers	2000	23	46000	3
	Water Cooler	1550	5	7750	5
	Water Purifier	35	5	175	5
	Refrigerator	210	1	210	1
	Freezer	1800	1	1800	1
	Grinder	750	1	750	1
	Exhaust	300	3	900	3
	Dough Machine	746	1	746	1

	TV	180	1	180	0

4. **BOYS HOSTEL 3**

AREA	LOAD	WATTAGE	CONNECTED LOAD QTY.	TOTAL LOAD CONNECTED	LOAD IN USE QTY.
BOYS HOSTEL 3	Tubelight	36	390	14040	305
	Fans	60	365	21900	294
	LEDs	9	45	405	41
	Geysers	2000	23	46000	4
	Water Cooler	1550	5	7750	4
	Water Purifier	35	5	175	5
	Refrigerator	210	1	210	1
	Freezer	1800	1	1800	1
	Grinder	750	1	750	1
	Exhaust	300	3	900	3
	Dough Machine	746	1	746	1
	TV	180	1	180	0

5. **TOTAL HOSTEL LOAD AND ITS CURERNT USEAGE.**

AREA	LOAD	WATTAGE	CONNECTED LOAD QTY.	TOTAL LOAD CONNECTED (W)	LOAD IN USE (W)
TOTAL HOSTEL LOAD	Tubelight	36	1318	47448	38124
	Fans	60	1165	69900	58740
	LEDs	9	236	2124	1566
	Geysers	2000	83	166000	22000
	Water Cooler	1550	18	27900	26350
	Water Purifier	35	18	630	630
	Refrigerator	210	4	840	840
	Freezer	1800	4	7200	7200
	Grinder	750	4	3000	3000
	Exhaust	300	12	3600	3600
	Dough Machine	746	4	2984	2984
	TV	180	4	720	0
	TOTAL		2870	332346	165034

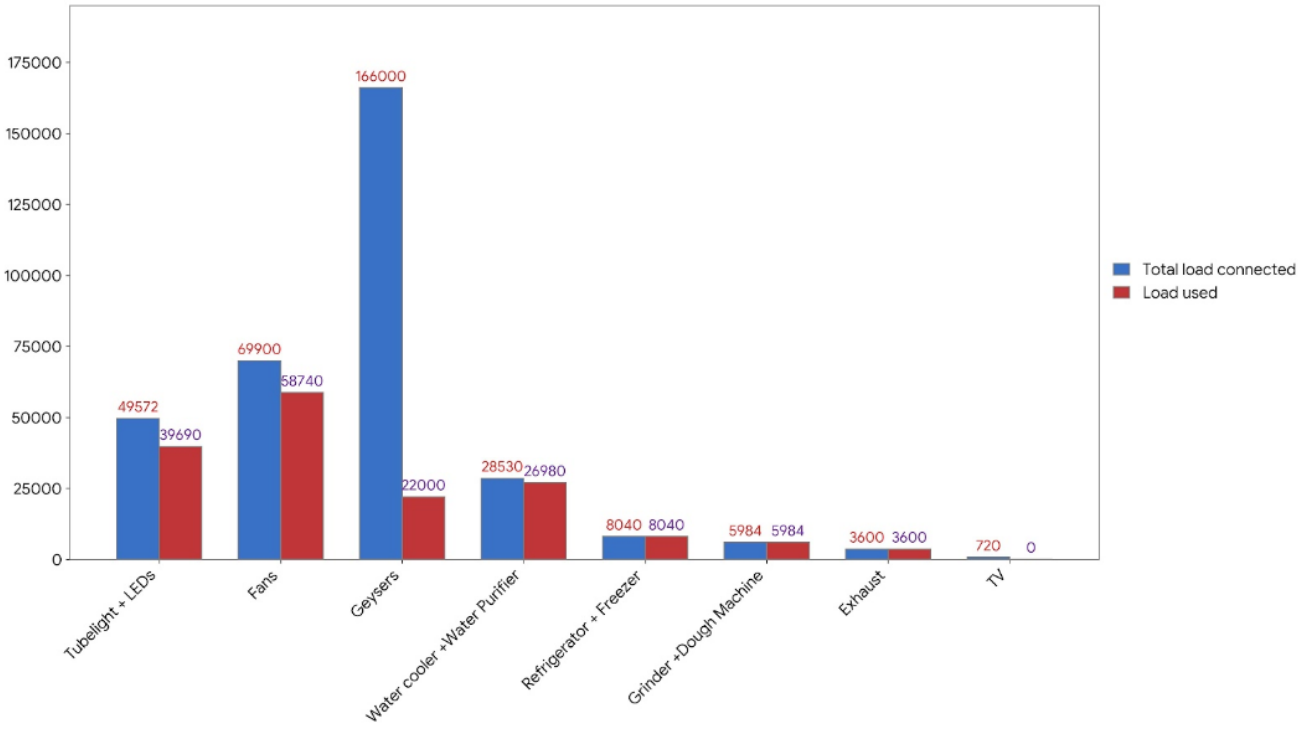


Fig 4.6: Bar graph of total connected load and load in use

In this bar graph we represented the connected loads as the orange bar and the green bar as the load which are being used in the month of February 2023. As we can see hostels are consuming the most amount of energy and classrooms consuming the least.

ANNUAL CONSUMPTION OF POWER OF THE YEAR 2025-26

Month & Year	Units Consumed (kWh)	Bill (Rs.)
Jan-25	45600	387600
Feb-25	48180	414630
Mar-25	79000	671500
Apr-25	134665	1101237
May-25	110680	901575
Jun-25	103775	797042
Jul-25	105130	855982
Aug-25	154030	1317378
Sep-25	157940	1339468
Oct-25	111385	947808
Nov-25	88665	776565
Dec-25	77235	698493
Jan-26	100785	913606
Total consumption	**1317070**	**11122884**
Average monthly Consumption	**109755.83**	**926907**

Annual Monthly Consumption by the institute

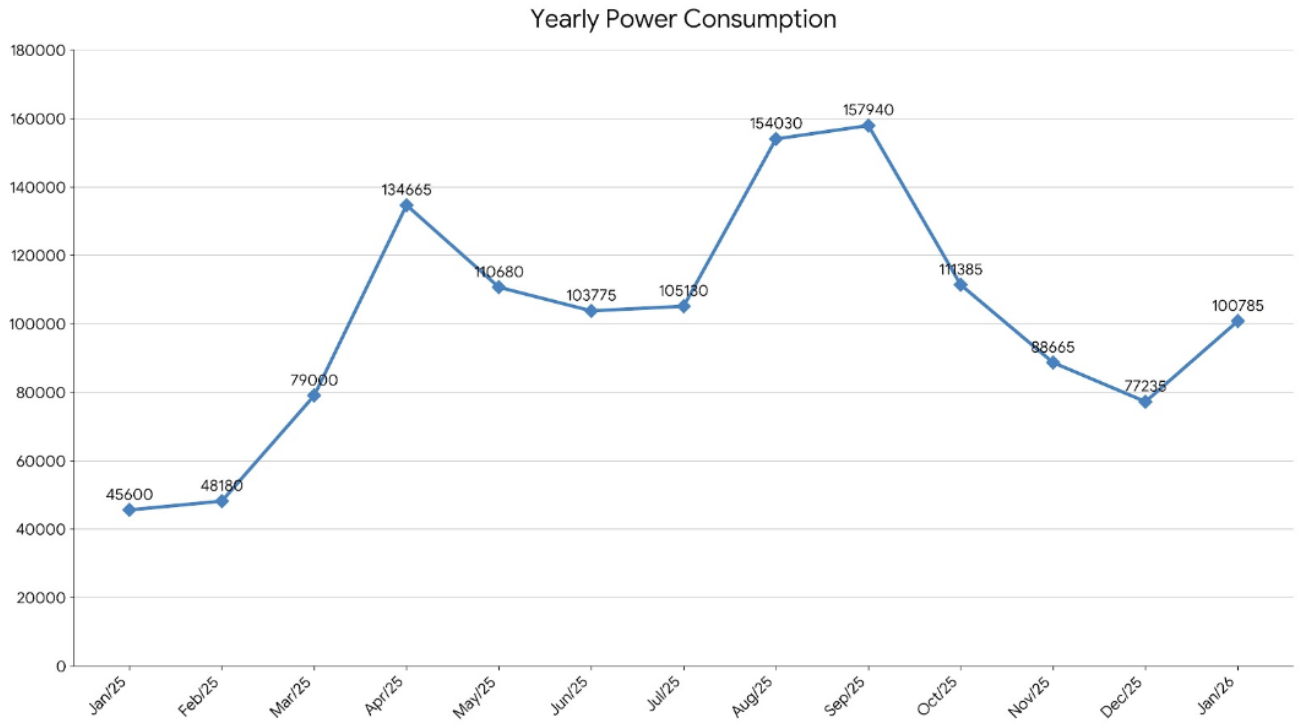


Fig 4.7: line graph of monthly electricity consumption

Now we calculated the units consumed by the particular appliances in January 2026

Appliances	kWh in a day (Jan)	kWh in Jan
Tube light	609.3	16281.3
Fans	0	0
AC	0	0
Table Fans	0	0
Geyser	1554	45384
Computer	298.46	9258.3
Kitchen Appliances	98.29	3041
Water cooler	680.7	20101.7
Printer + Projector	13.28	395.5
LAB Apparatus	25.584	793.104
LEDs	23.258	725.16
Ground + Street Light	155	4805
Total	**3457.872**	**100785.064**

Units Consumed by every appliance in Jan 2026

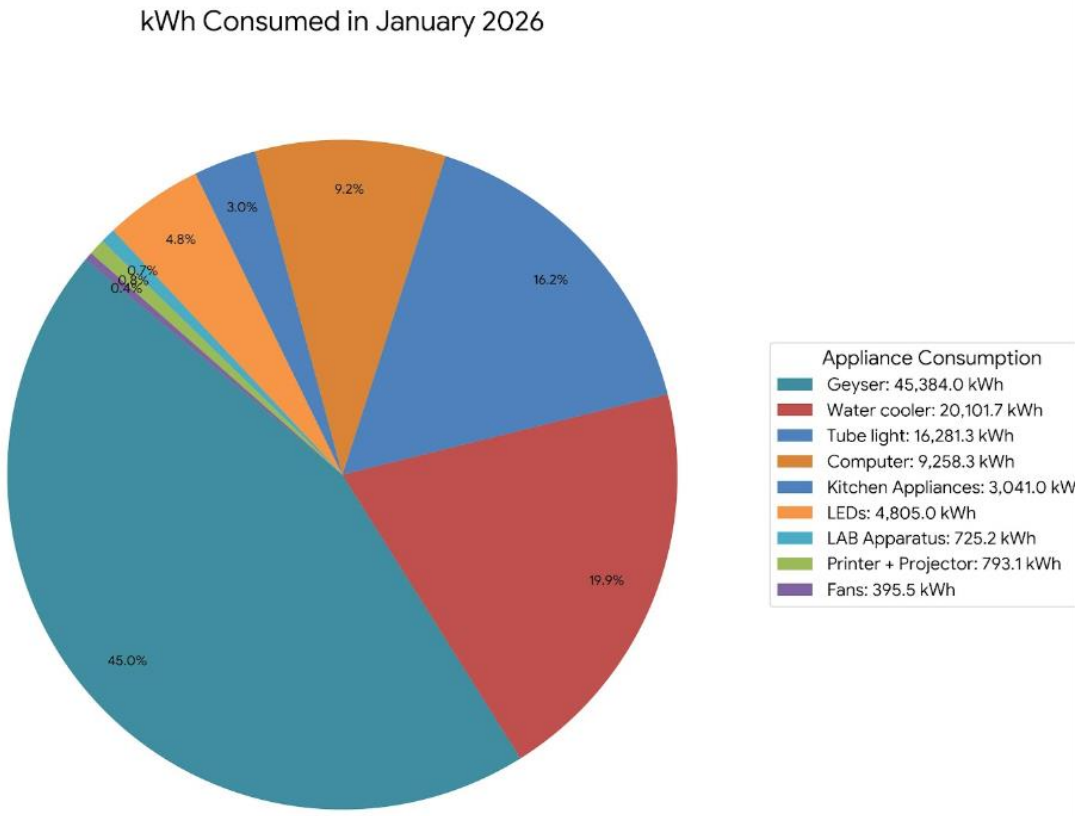


Fig 4.8: Pie chart of units consumed by electrical appliances in Jan 2026

In this Pie chart we represented the units consumed in kWh by every appliance in the college in February 2023 so the geysers consume the most units and the fans, AC and the Table Fans are not being used in that month so it consumed no units after that Printers and Projectors consumed least amount of energy.

Group 4 Guest House

Month	Units Consumed
Jan-25	5818.81
Feb-25	5818.81
Mar-25	5818.81
Apr-25	6714
May-25	6714
Jun-25	6714
Jul-25	6714
Aug-25	6714
Sep-25	6714
Oct-25	6714
Nov-25	5818.81
Dec-25	5818.81
Jan-26	5818.81

D. **Group 5** Street Lights, Court Lights

Total street lights = 96 of 60 watt

Total court lights = 62 of 200 watt

Watt Unit Consumption Calculation of Street Lights

96 lights (60 watt) = 96 × 60 = 5760 watt

Now, the energy consumption in 13 hr = 13 × 5760 watt = 74880 Whour

Energy usage in one month (that is 30 days) = 30 × 13 × 5760 watt = 2246400 Whour

Total usage is 2246400 Whour

Knowing the fact that 1 KWh = 1000 Whour

Therefore, total consumption in kWh = 2246400 / 1000 = 2246.4 kWh

Also 1 unit = 1 kWh

Therefore, total energy consumption in one month = 2246.4 Unit

We also know that 1 unit is equivalent to 1 kWh

Therefore, total energy used in a month = 2246.4 Unit

Unit Consumption Calculation of Court Light

62 lights (200 watt) = 62 × 200 = 12400 Watt

Now, the energy consumption in 4 hr = 4 × 12400 Whour = 49600 Whour

Energy consumption in one month (that is 30 days) = 30 × 4 × 12400 Whour

Total Usage = 1488000 Whour

We know that 1 Kwh = 1000 Whour

Therefore, total consumption in kwh = 1488000 / 1000 = 1454.7 Kwh

We also know that 1 unit equals 1 kwh

Therefore, Total energy consumption in one Month = 1488 Unit

E. **Group 6** Guard Room, Security, Gail, General store, Canteen, ATM, Nescafe.

Month	Units Consumed in kWh						
	Guard Room	Security	General store	Canteen	ATM	Gail India	Nescafe
Jan-25	489	582	30	136	120	764	370
Feb-25	616	450	0	423	120	433	591
Mar-25	265	419	0	0	114	425	434
Apr-25	316	510	0	1175	119	888	713
May-25	354	547	0	0	121	512	881
Jun-25	447	596	0	436	126	1688	458
Jul-25	485	622	0	0	121	528	354
Aug-25	479	653	0	1221	126	1065	547
Sep-25	460	578	0	956	127	982	1016
Oct-25	342	580	1286	0	65	1472	708
Nov-25	287	477	273	614	44	394	520
Dec-25	396	487	190	424	40	434	513
Jan-26	378	636	0	248	42	536	342
Total	**5314**	**7137**	**1779**	**5633**	**1285**	**10121**	**7447**

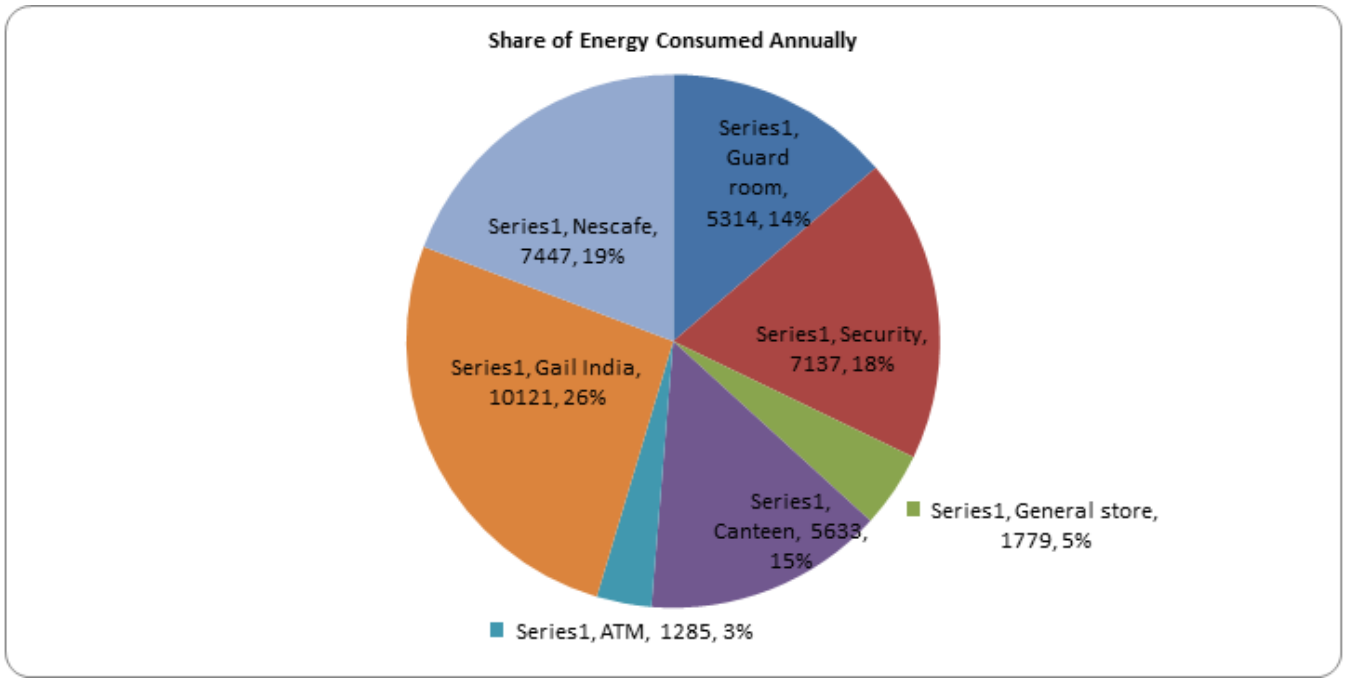
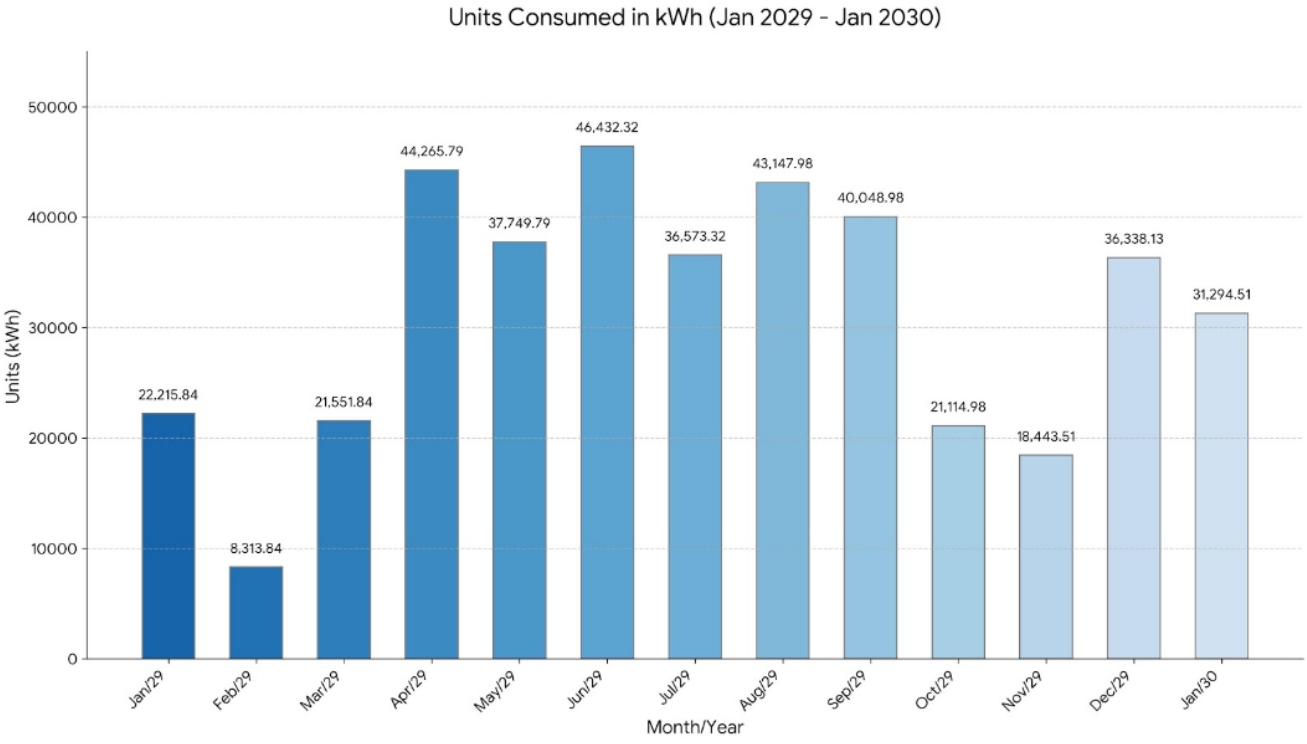


Fig 4.9 Share of energy consumed annually by group 6

F. **Group 7** Academic Block

Month	Units Consumed in Academic Block (kWh)
Jan-25	22215.84
Feb-25	8313.84
Mar-25	21551.84
Apr-25	44265.79
May-25	37749.79
Jun-25	46432.32
Jul-25	36573.32
Aug-25	43147.98
Sep-25	40048.98
Oct-25	21114.98
Nov-25	18443.51
Dec-25	36338.13
Jan-26	31294.51



4.10 Units consumed month wise by academic block

CHAPTER 5: Implementation Measures

5.1 Replacing Street Lights to Solar Street Lights.

Working of solar street lights:



Fig 5.1 Working of solar street lights

A solar cell consists of layers of negatively-charged electrons and positively charged spaces. The electrons in these solar cells are triggered and forced into the positive spaces by sunlight. This creates an electrical circuit, which produces direct current electricity as the electrons flow across it. A solar panel is made up of several solar cells, each of which is formed of the semiconductor material known as crystalline silicone. A charge controller connects the battery and solar panel. Charge controllers work to safeguard batteries by controlling voltage and stopping current from flowing backward through solar cells when charging isn't occurring.

Both the sunlight-to-current conversion mechanism and solar cell power generation cease at dusk. When the solar lamp's photoreceptors detect a decrease in sunshine, they begin to supply the light source with battery-stored energy. Once more, the photoreceptors automatically turned off the light at daybreak.

Calculation:

Total energy consumption in one Month by street light = 2246.4 Unit

Tariff @ Rs. 6.55(after taking avg. of 2H1 and 3H1)

Cost of energy per month = 2246.4 * Rs. 6.55 = Rs.14713.92 4

Cost of energy per year = Rs.176567.04

Specification and costing of solar street light:

42Ah Battery

110w Solar panel

50w LED

Cost of integrated system = Rs.9000

Cost of 96 street light = 9000*96 = Rs.864000

Payback period = 58 months

Advantages of solar street lights.

Solar street lighting systems use solar panels to generate electricity during the day and store it in batteries. This stored energy is then used to power lights at night.

Advantages include:

- No dependency on grid electricity
- Reduced electricity bills
- Low maintenance requirements

Although the initial investment is higher, the long-term savings make it a practical solution.

Replacing water heaters by water heating system running on solar power

Working of water heating system running on solar power:

Sunlight reaches the collector panel, which is a major component of the solar powered water heating system. Solar thermal energy is used by the absorber, a black absorbing surface inside the collectors, and is then transferred to the water that passes past it. The hot water is gathered in an insulated tank to stop heat loss. The thermosiphon system allows water to automatically flow from the tank via the collectors and back to the same. There are two variations of same based on the collector system: Hot water is kept in an insulated storage tank being a solar heater, which uses a collector to gather solar energy. You can utilize the hot water that has been kept whenever you choose.

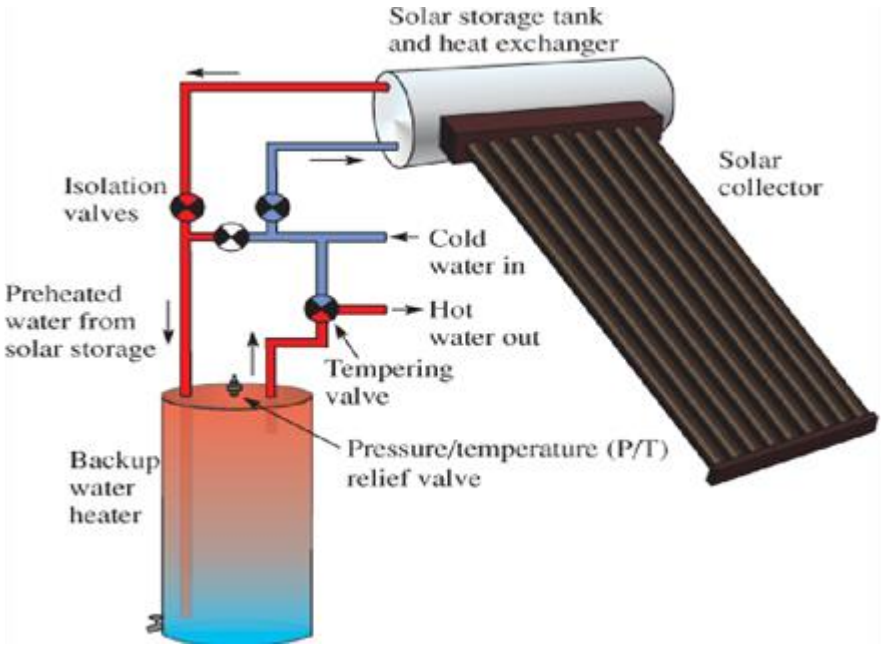


Fig 5.2 Working of solar water heating system

Calculation:

Hot waters used in hostels = 600

Assuming amount of hot water per student = 20 litres / day.

Accumulated water used by hostel per day =12000 litres.

Typically, 2 m2 ogives 125 Litres per day at 60 °C

Normal hot water temperature is to be 40 °C.

However, since piping losses are also involved, we will assume that the user uses hot water at 60 °C. This means that the hostel will need 192 m2 for collector area. One can choose to use natural circulation or forced circulation.

Proper pump control and a variety of accessories are necessary for forced circulation. As a result, 48 m2 of collector area is needed for each of the four hostels when choosing a natural circulation system.

Cost for collector roughly sums uo to Rs. 10,000 / m2 .

Total capital cost for hostel = 10,000*106 = Rs 19,20,000

Total energy used by water heater (hostels) = 472 unit/day

Consumption over a year = 70800 unit (150 working days)

Cost of electricity taken at Rs. 6.55 / Unit ,

Total electricity bill of heating = 70800*6.55 = Rs. 4,63,740 /-
Payback Period (SPP) = (capital cost / bill per year) = 4 years. This being on the higher side due to the lower strata of students in hostels.

General Recommendations

From what was observed during the audit, a few practical changes can be suggested which do not require major modification to the existing system but can still improve overall energy usage.

First of all, it would be relatively simple to upgrade the lighting fixtures. Currently, a lot of different fixtures operate using traditional light bulbs, while they could be replaced by LED lights gradually without much hassle. In addition to the energy savings achieved by switching from the traditional bulbs, LEDs tend to operate for longer, which means that they will require fewer replacements.

Maintenance seems like another thing that should be addressed. While many people do not realize it, poorly-maintained electrical systems usually waste extra energy. Loose connections, aged wiring, or simply unmaintained equipment often lead to increased consumption, which may seem strange at first sight.

Another thing that might be worth paying attention to is high-power equipment. Water heaters and pumps consume quite a bit of power, so making sure that their usage is minimized whenever possible is crucial. Avoiding unnecessary usage of such appliances is one way to make them work properly and save extra power.

It became clear that the users' actions affect the energy consumption as well. The more people use the equipment incorrectly, the harder the system needs to work. Educating the students and the staff about the correct way to use it might be helpful here.

Finally, some alternative energy sources might be considered. Using solar energy for street lighting and water heaters is just one possible solution. Even if it requires additional investments initially, its benefits in the long term are obvious.

At the same time, this audit cannot be considered a one-time deal. It would be wise to repeat it after a certain period of time to see whether changes made earlier helped to achieve the goals set.

All things considered, those suggestions are relatively simple to implement but effective anyway.

CHAPTER 6: RESULTS AND DISCUSSION

5.1 Annual Energy Consumption Results

A total consumption of electricity in the period January 2025-January 2026 in BIT Mesra Off-Campus Patna has been found to be 1,317,070 kWh amounting to an approximate cost of ₹1.11 crores. An average consumption of electricity in a month has been found to be 109,755.83 kWh, i.e., approximately ₹9.27 lakhs. The electricity supply for the campus is provided through two transformers with a rating of 200 KVA and 450 KVA respectively. Important results based on categories of loads have been described below.

Pump Room: The 15HP and 5HP water pumps consumed 4,356 and 1,454.7 units per month whereas the 20HP water pump used in peak summer months consumes 6,714 units per month.

Faculty Quarters: The monthly consumption varies from 4,100 kWh in March to 14,706 kWh in June which reflects that the monthly consumption has direct dependency on temperature and air conditioning.

Hostels (maximum consumer of electricity): The maximum connected load for the hostels is 332,346 W and actual load in use is 165,034 W resulting into a load factor of 50%. There are geysers with a total load of 166,000 W in hostel, accounting for about 50% of its capacity.

Academic Block: The peak demand in the academic block has been seen in June (exam season) where it was 46,432 kWh whereas there was minimum demand in February when it was 8,313 kWh.

Street and Court Lights: There are 96 lights (each having a rating of 60 W) consuming 2,246 units per month whereas 62 lights in the courts (200 W each) consume 1,488 units per month.

5.2 Discussion

Preponderance of Thermal Loads

Geyser use and water coolers together contribute to approximately 65% of total energy usage during peak winter periods. Therefore, water heating becomes the highest priority energy-saving activity, thus providing the strongest justification for solar water heating installation.

Factor of Load and Rationale

Given the 50% load factor of the hostels, we can conclude that one-half of the installed power capacity lies dormant at all times. Timer-controlled geysers, occupancy sensors for fans and exhausts will yield instant savings with zero cost investment.

Other Measures

Roof cooling paint, greenery, LED retrofitting of remaining tubular bulbs, and exhaust fan timers could be considered likely candidates for a supplementary saving of 10-20% without much capital expense.

5.3 Key Takeaways from Energy Audit Report

Total annual electricity usage: 1,317,070 kWh, with total cost of ₹1.11 crore (₹9.27 lakh per month).

Pre-dominance of hostels; thermal loads (geysers + coolers) make up roughly 65% of winter electricity consumption.

There is significant seasonality; peaks are created by different factors for summer (cooling) and winter (heating), calling for seasonal interventions.

Solar water heating installation (simple payback of 4 years; annual saving of ₹4.63 lakh) is the most financially rewarding intervention possible.

Solar street lighting (payback of roughly 58 months) renders outdoor lighting independent of electricity grid supply post-payback.

Inexpensive interventions (LED retrofits, timers, cool roofs) could potentially save another 10-20%.

Full-scale implementation will turn BIT Mesra Patna into the role model of energy efficiency among Bihar universities and colleges.

CHAPTER 7: CONCLUSION AND FUTURE WORK

7.1 Conclusion

The conducted energy audit gives an accurate picture of the distribution of electrical energy utilization within various parts of the BIT Mesra Off-Campus, Patna. While the consumption value might appear vague, the division of the total electrical consumption value into distinct parts allows for a deeper understanding of where energy is being consumed.

It can be observed from the energy audit report that the hostels consume the most electricity in the BIT Mesra Off-Campus, Patna. The reason behind this is that these structures have consistent occupancy and utilize numerous devices, including lighting fixtures, fans, and water heating units. Water heating proved to be among the leading factors influencing the amount of energy consumed by hostels.

Regarding academic structures, it can be stated that the consumption pattern of electricity in these buildings varies depending on time. The street lighting installation has a uniform electricity consumption pattern. It is also worth noting that in many cases, the capacity of electrical equipment exceeds actual utilization, which implies that energy could be saved through proper management.

One of the key conclusions of the energy audit conducted for BIT Mesra Off-Campus, Patna is that the improvement of energy efficiency can be achieved through simple means. For instance, using LED lights, decreasing the frequency of electricity utilization, and regular maintenance of equipment will help improve the situation. Solutions that allow replacing conventional power sources include the introduction of solar-powered street lighting and water heating facilities.

To summarize, it can be concluded that there is considerable potential for improving the energy consumption structure in BIT Mesra Off-Campus, Patna, allowing for saving costs and decreasing ecological footprint without interfering with the normal functioning of the institution.

7.2 Future Work

Even though the current audit gives good insights, there is much to do in terms of improving energy management in this college.

First, an installation of real-time energy monitoring system will provide additional data that could be used to detect any changes, including abnormal increases in energy consumption. The ability to have this information will make it possible to respond promptly.

Second, energy audits should focus on automation systems. Lighting, air conditioning, heating systems, etc., could be optimized by installing automatic controls. For example, timers, motion sensors, and lighting controls can be installed in some buildings.

Third, future studies could be aimed at conducting seasonal analysis since the level of energy consumption depends on weather conditions. It could be done by installing appropriate monitoring devices.

Fourth, a number of campaigns could be started to increase the users' awareness of how their energy consumption impacts total costs. It should help to change some habits and practices associated with energy consumption.

Fifth, in addition to solar energy, other sources can be researched and analyzed to determine whether they could be applied to increase efficiency. In other words, other energy sources could be researched.

Sixth, audits could be repeated regularly to ensure continuous improvement. As already stated, energy audits are an ongoing process, which means that the results of each audit should be considered when implementing changes and improving energy management system.

To conclude, there is a lot to do in order to further extend this work.

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